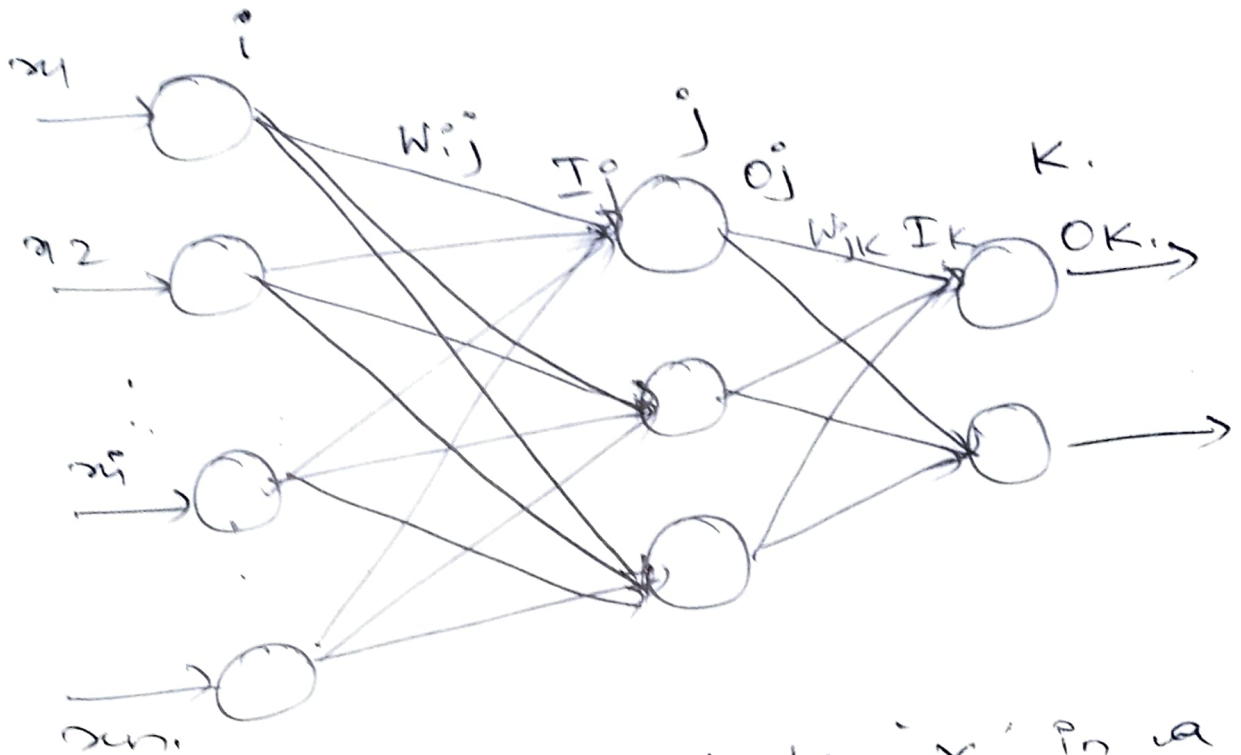


Multi-layer Perceptron

- * Layer / Network of Perceptron units attached to solve a linearly in-separable problem.
- * Mostly fully Connected Network
- * Named as two-layer ^{w/w} if it has three layers [Input layer - hidden layer - Output layer]
- * Named as three layer network if it has four layers [Input layer - hidden layer 1 - hidden layer 2 - output layer]
- * Activation function used in each unit of hidden / output layer may be logistic / sigmoidal function
- * Sigmoidal is applicable for Binary class classification model.
- * tanh (hyperbolic tangent) function can also be used

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Multi-layer Perceptron - Feed Forward Neural Network & Backpropagation ①



For each training tuple 'x' in a Dataset 'D'

for each Input layer unit j

$$O_j^o = I_j^o$$

For each hidden/output unit layer j

$$I_j^o = \sum_i W_{ij}^{oo} O_i^o + \theta_j^o$$

$$O_j^o = \frac{1}{1 + e^{-I_j^o}}$$

Back propagate the error δ for each unit j in the output layer

$$Err_j^o = O_j^o (1 - O_j^o) (T_j^o - O_j^o)$$

②

T_j = Target output
 O_j = Desired output.

for each unit 'j' in the hidden layer from last to first hidden layer

$$Error_j = O_j(1 - O_j) \sum_k Error_k W_{jk}$$

for each weight W_{ij} in Network

$$W_{ij} = W_{ij} + \lambda Error_j O_i$$

for each bias θ_j

$$\theta_j = \theta_j + \lambda Error_j$$

where ' λ ' is a learning rate

Questions:

Design a MLP with Back propagation to implement a OR Gate. Assume data wherever necessary.